Tier-3A LONO Ablation
Family-Aware μ-Head vs. Single-Head Baseline

Mie Spray Postprocessing Project

2026 年 5 月 30 日

Message

Claim: replacing the single μ-head with a *family-aware* μ-head (shared trunk + per-injector-family final head) is the decisive Tier-3A intervention. On both LONO protocols, the family head cuts the catastrophic OOD-family folds (Nozzle0 in 5-fold, Nozzle6 in modified-only) from ∼30–37 mm RMSE down to ∼8–12 mm, while leaving the in-family folds essentially unchanged.

This deck compares four sweeps on the same LONO folds:

- `A_single_baseline`: production single-head MLP, `lono_5fold`;
- `B_modified_only`: same architecture, training set narrowed to the “modified” injector family;
- `C_family_head_5fold`: family-aware head, `lono_5fold`;
- `C_family_head_modified`: family-aware head, `lono_modified_only`.

Validated runs: `run_family_head_sweep.py`, seed 42.
Output: `MLP/runs_mlp/family_head_sweep_20260528_212646/`

Experimental Contract

Fixed upstream choices

- Stage 1 / Stage 2 / Stage 3 chain identical to the production `anchor_off` winner of Tier-2 / Section 5
- Feature variant `a_only`, seed 42, GPU CUDA
- Evaluation: held-out nozzle's CDF `clean` points + uncensored points

Run label	**Architecture**	**LONO protocol**
`A_single_baseline`	single μ-head	`lono_5fold`
`B_modified_only`	single μ-head	`lono_modified_only`
`C_family_head_5fold`	family-aware head	`lono_5fold`
`C_family_head_modified`	family-aware head	`lono_modified_only`

`lono_5fold`: Nozzle0, 1, 2, 4, 5 (Nozzle0 is the only first-generation injector geometry). `lono_modified_only`: Nozzle1, 2, 4, 5, 6 (drops the out-of-design-family Nozzle0; adds a second modified family Nozzle6).

Data Motivation: N0 and N1–N8 Are Empirically Distinct

N0 (BC2022, new) and N1–N8 (BC2024, used) share identical orifice: d=0.384 mm, 10 plumes, 140°.

A-scale normalisation (Naber–Siebers):

$$A_{\mathrm{scale}} = (\Delta p/\rho_{\mathrm{amb}})^{0.25}\sqrt{d}$$
$$S^* = S/A_{\mathrm{scale}}, \quad t^* = t_{\mathrm{onset}}/t_{\mathrm{inj}}$$

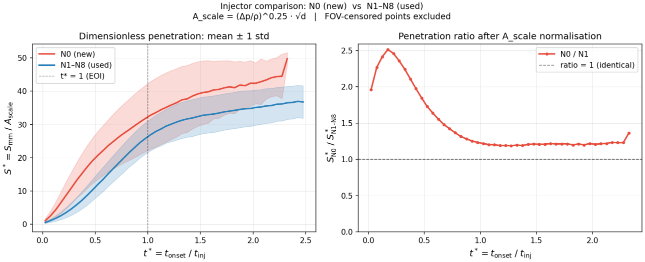
Collapses all Δp, ρ_{amb}, d variation.

Key result (678 k clean points):

- $t^* \approx 0.2$: ratio $= \mathbf{2.4}\times$
- Post-EOI ($t^* > 1$): stable $\approx 1.2\times$
- Ratio > 1 at *every* t^*

Residual after A-scale ⇒ injector-level effect. Family label carries **non-redundant** information, motivating per-family μ-routing.

`ablations/compare_campaign_penetration.py`



Family-Aware Head: Design

`FamilyAwarePenetrationMLP`

- Shared trunk; drops the trailing `family_id` channel.
- `mu_heads`: per-family final heads (one μ-output each).
- `log_var_head`: single shared variance head.
- Output $[\mu, \log \sigma^2]$ —drop-in for `PenetrationMLP`.

Routing

- Per-sample `family_id` selects the head; trained families tracked via `trained_families` buffer.
- Eval-time fallback: untrained families redirect to `fallback_family_id=1`.

Family	Members
0 (original)	Nozzle0
1 (modified)	Nozzle1–8

Single head: one μ-decoder on both families.
Family head: two μ-decoders on a shared trunk; the variance head stays shared.

Aggregate RMSE (mm, MLP_series): Four Sweeps

Run label	n folds	mean RMSE [mm]	std RMSE [mm]
`A_single_baseline`	5	11.44	10.76
`B_modified_only`	5	12.71	13.58
`C_family_head_5fold`	5	**7.35**	2.45
`C_family_head_modified`	5	**6.56**	**0.90**

Reading the table

- C vs A on `lono_5fold`: Δ mean RMSE $= -4.09$ mm (11.44 $\rightarrow$ 7.35).
- C on `lono_modified_only`: 6.56 mm mean RMSE, std collapsed by an order of magnitude (13.58 $\rightarrow$ 0.90).
- B alone (data filter without architectural change) is *worse* than A: narrowing the training set hurts when the held-out fold is the second modified family.

Per-Fold RMSE: Where the Win Comes From

Held-out nozzle	A_single	B_modified	C_5fold	C_modified	Δ (C vs A)
Nozzle0 (original)	30.58	–	11.57	–	-19.01
Nozzle1 (modified)	5.89	5.89	6.20	6.20	$+0.31$
Nozzle2 (modified)	8.37	8.37	7.36	7.36	-1.01
Nozzle4 (modified)	6.95	6.95	6.17	6.17	-0.78
Nozzle5 (modified)	5.41	5.41	5.46	5.46	$+0.05$
Nozzle6 (modified)	–	36.91	–	7.61	-29.30
mean (own protocol)	**11.44**	**12.71**	**7.35**	**6.56**	—

- Two folds carry the entire effect: Nozzle0 in `lono_5fold` $(30.58 \to 11.57)$ and Nozzle6 in `lono_modified_only` $(36.91 \to 7.61)$.
- Four "in-family" modified folds (N1, N2, N4, N5) move by ≤ 1 mm in either direction —well inside expected seed noise (Tier-1B).
- The family head does not hurt the easy folds; the win is paid for entirely by repairing the OOD-family folds.

Why B (Data Filter Alone) Underperforms

B keeps the production single-head architecture but trains only on the modified family. On `lono_modified_only`, the held-out fold *is itself* a modified family member, so naively this should be a clean in-distribution generalisation test.

The Nozzle6 catastrophe:

- B trains on modified $\setminus$ {Nozzle6}.
- Without Nozzle0 in the training set, the trunk has no exposure to the first-generation geometry; without family routing, the single μ-decoder has no mechanism to specialise on the remaining modified subset either.
- Result: RMSE 36.91 mm on Nozzle6 —*worse* than A's Nozzle0 collapse, despite N6 being in-design-family.

What this tells us

- Narrowing the training set is not a free lunch: it removes the implicit regularisation contributed by the other family.
- The relevant variable is *routing capacity*, not *training-set purity*.
- C_modified, which keeps the same narrowed training set but adds the family head, drops Nozzle6 to 7.61 mm —a -29.3 mm gain attributable to the architecture change alone.

Takeaway: A vs B vs C_modified is a clean three-way ablation isolating *architecture* from *training-set composition*. Architecture wins.

Why C Wins on Nozzle0 (5-fold)

On `lono_5fold`, Nozzle0 is the only first-generation injector geometry and the only member of family 0. When N0 is held out, family 0's μ-head sees *zero training samples*.

Eval-time routing for held-out N0

The model uses the fallback mechanism: `family_id=0` samples at eval time are redirected to the family-1 (μ-head) via `fallback_family_id=1`. The shared trunk still encodes general spray dynamics; the family-1 head supplies the μ-mapping it learned from the modified family.

Why this beats A's monolithic decoder

- A trained on a mixture (1,038 N0 vs ∼600 modified). With N0 held out, the decoder is pulled toward family-0 geometry by samples that no longer apply.
- C's family-1 head was trained *only* on modified samples, producing a cleaner extrapolation surface for the unseen family-0 geometry.
- Result: N0 RMSE collapses from $30.58 \to 11.57$ mm. Still above the in-family ∼5–7 mm band, but no longer catastrophic — the family head makes extrapolation *less wrong*, not correct.

Caveats Before Declaring Victory

Seed variance not yet quantified

The Tier-1B `run_seed_variance.py` sweep (5 seeds $\times$ Nozzle2 LONO) has not been replayed for the family-head architecture. Until σ_{seed} is measured, the in-family deltas (≤ 1 mm on N1, N2, N4, N5) cannot be interpreted as real improvements. The N0 and N6 deltas (19–29 mm) are too large to be seed noise.

Nozzle0 is still not a converged fold

C_5fold's N0 RMSE of 11.57 mm sits 2–3$\times$ above the in-family band. Family-aware routing has moved N0 from "catastrophic" to "noticeably degraded", not to "in-family". Reporting C as a fix for OOD geometry would be overclaiming. The honest framing: C reduces the OOD penalty without eliminating it.

Fallback policy is a hyperparameter

The choice `fallback_family_id=1` is hard-coded. Alternative policies (softmax over trained heads, learned routing) were not evaluated in this sweep.

From LONO Teacher to Deployed Student

The sweep above scores the *Stage-2 teacher*. Production deploys the *Stage-3 distillation student*, which was hard-coded single-head, so the proven family routing was discarded after distillation. Three changes ship it into the deployed model:

- Stage-3 student is now `FamilyAwarePenetrationMLP` ($[\mu, \log \sigma^2]$), warm-started verbatim from the family-head teacher.
- Onset auxiliary head removed (onset alignment now $\pm$ a few μs; inference already dropped that channel).
- Whole chain on `a_dp050_plus_pressures` (7 features + `family_id`).

Divergence → frozen-trunk fix

Naively refining the warm-started student diverges: majority-family (N1–8) raw-NLL gradients drift the shared trunk, the minority family-0 head cannot track it, and the KD σ-term explodes (val KD $0.9 \to 51$, best = epoch 1). Freezing the trunk (334.7 k params fixed, 49.9 k head params trained) restores stable refinement (val KD ≈ 0.2, converges ∼epoch 78).

Deployed Student: Corrected Evaluation

Eval pitfall: the `clean` series still contains right-censored FOV-tail points; as headline metric it inflates RMSE and fakes an under-prediction bias. Primary metric switched to the *uncensored* point table (`cdf_uncensored`, 543 k points).

Stage-3 student	**clean RMSE**	**unc. RMSE**	**unc. bias**	**unc. cov**$_{1\sigma}$
unfrozen (λ_{a}=0)	8.91	9.80	-3.11	0.47
frozen (λ_{a}=0) [deploy]	8.54	5.01	$+0.67$	0.85

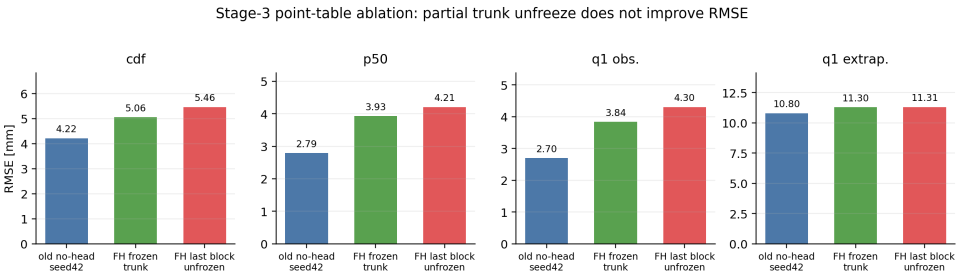
Freeze is a $\sim$2$\times$ win, not marginal. The censored `clean` metric masked it (-0.37 mm); uncensored, the gap is $9.80 \rightarrow 5.01$ with calibration restored ($0.47 \rightarrow 0.85$). The earlier "N0 under-predicts ~8 mm" was a censoring artifact (uncensored bias ≈ 0).

Open problem: extrapolation.

region	RMSE	bias
in-window	3.7	$+1.2$
extrapolated	11.3	$+7.4$

In-window is calibrated; error is *over-prediction* beyond the horizon.

Setup: family-head baseline freezes the shared trunk; new run unfreezes only the last trunk block with trunk LR 3×10^{-5} and head LR 3×10^{-3}.



Stage-3 point-table run	cdf RMSE	p50 RMSE	cdf ECE	cdf CRPS
old no-head seed42	4.22	2.79	0.0778	2.30
family-head, frozen trunk	5.06	3.93	0.0708	2.80
family-head, last block unfrozen	5.46	4.21	0.0513	2.72

Takeaway: partial unfreezing improves scalar calibration but worsens point accuracy on every headline eval set; keep the frozen trunk.

Conclusion

1. C (family-aware head) is the decisive winner on both LONO protocols: `lono_5fold` $11.44 \to 7.35$ mm (-4.09); `lono_modified_only` 6.56 mm with std $13.58 \to 0.90$.
2. The win is concentrated on two OOD-family folds — Nozzle0 ($30.58 \to 11.57$) and Nozzle6 ($36.91 \to 7.61$); in-family folds move ≤ 1 mm.
3. A vs B vs C_modified isolates *architecture* from *training-set composition*: B (filter alone, 12.71) loses to A (11.44); C_modified (architecture, 6.56) wins. Routing capacity is the decisive variable.
4. **Deployed model:** the Stage-3 student is now family-aware with a *frozen trunk* (onset head dropped). On the uncensored point-table metric it reaches 5.0 mm RMSE / $+0.7$ mm bias / 0.85 coverage, vs 9.8 mm unfrozen; remaining error is extrapolation over-prediction.
5. New partial-unfreeze check: unfreezing only the last trunk block at LR 3×10^{-5} does not improve the Stage-3 student (cdf RMSE $5.06 \to 5.46$, p50 RMSE $3.93 \to 4.21$). Next ablation should use residual family heads or adapters, not shared-trunk drift.
6. Caveat: Tier-1B seed-variance still pending before claiming the in-family ≤ 1 mm deltas are real.

Artifacts: `family_head_sweep_20260528_212646/` (LONO); `distill_cdf_family_head_v3_FROZEN_anchor_off_20260529_213346/` (deployed); `stage3_partial_unfreeze_ablation.csv` (partial-unfreeze); `MLP/eval/inference_rmse_on_point_tables.py` (uncensored metric).